

José Francisco Urdaneta

AI Technical Lead · Agentic AI & LLM Architecture

Multi-Agent Systems · RAG · Architectural Governance · Distributed Systems

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PROFILE

Engineering leader with 20+ years in software engineering and technical leadership, currently focused on agentic AI and LLM architecture. I directed Pulse AI, a production agentic platform for regulated financial and insurance domains multi-agent orchestration, multi-engine LLM integration, RAG and SLM strategies, and an inferential rules layer for traceable decision making. I operate at architectural governance level, owning architecture and infrastructure decisions, governing cost and resource provisioning, and aligning engineering teams and stakeholders, while remaining hands-on: I build and ship my own AI products on Google Cloud Vertex AI with Python and React.

TECHNICAL SKILLS

AI & LLM: Agentic AI · Multi-agent orchestration · LLM integration (OpenAI, Gemini, DeepSeek, Anthropic) · RAG pipelines · SLM strategies · Prompt engineering · Fine-tuning / model training · AI cost estimation & optimization

Cloud & AI Platforms: Google Cloud Platform · Vertex AI (model deployment, grounding, managed AI services) · Firestore · Distributed systems · Cloud architecture & resource governance

Engineering: Python · FastAPI · Java · React · Angular · Native Android / iOS · REST API design & integration · Drools (business rules engine) · Data lakes & analytics · Biometrics / OCR / e-Signature

Leadership & Process: Technical leadership of cross-functional teams · Agile (Scrum / Kanban) · Stakeholder management · Architectural decision negotiation · Production reliability & SLA governance

PROFESSIONAL EXPERIENCE

4Told FinTech Colombia (remote)

Apr 2017 – Present

Director of Operations (2020–Present) · Director of Digital Platforms (2018–2020) · Technical Lead (2017–2018)

B2B fintech: digital credit origination, onboarding and risk-decisioning platforms for retail and regulated financial institutions.

- Directed Pulse AI, 4Told's agentic AI platform for regulated finance and insurance: multi-agent architecture (receptionist agent, intelligent orchestrator, domain-specialist agents) with an inferential business-rules layer (Drools) governing every decision for regulatory traceability.
- Governed the multi-engine LLM architecture integrating OpenAI, Gemini and DeepSeek behind a single orchestration layer, defining routing strategy, contract abstraction and provider-independence at architectural level.
- Owned the cost/latency trade-off across the AI stack: designed RAG and SLM (small language model) strategies to migrate workloads to faster, cheaper models; estimated AI spend and governed compute resource provisioning.
- Architected the RAG layer with operational document ingestion (product terms, policies, conditions) feeding active retrieval and reasoning for the specialist agents.
- Defined governance controls separating generative conversation from business decisioning, enabling compliant inbound and outbound operation under consent and contact rules, a hard requirement in regulated markets.
- Integrated agents omnichannel across CRM, core banking, mobile banking, web and WhatsApp; defined a 12-week agile MVP roadmap for agentic pilots.
- Led a cross-functional engineering team of 20 (backend, web, mobile, rules engine, implementation, PM, technical support) under Scrum/Kanban across multiple countries and client-facing tiers.
- Scaled platform operations by 82% (453 to 825 operations/month) while cutting support tickets by 83% (208 to 36/month) in the same cycle, evidence of architectural and operational stability under growing throughput.
- Cut operating costs by 30% and support response times by 40%, delivered 25% more projects per year on the same budget, and drove 10 enterprise implementations across Colombia and Peru.

Vedana Solutions LatAm (remote)

Dec 2022 – Dec 2025

Technical Director (CTO)

- Designed and implemented data-lake analytics products (Orienta, Loanware), architecting ingestion, retrieval and AI-driven analysis to turn enterprise data into decision-making capability.
- Owned the full technical product cycle: conceptualization, architecture, development kickoff, UAT, risk analysis and go-to-market, negotiating technical scope directly with stakeholders.

Mobile Development Manager (2014–2016) · BPM Development Coordinator (2010–2014)

- Launched the bank's transactional digital platform (web and native mobile) into production: 5,000 daily active users and USD 2M/day transacted, 20% more operations than the legacy system, running high-throughput distributed systems in a regulated environment.
- Led a cross-functional team of 15; shipped to Play Store and App Store under enforced quality standards.

Earlier experience

Pyramid Consulting (Madrid), Project Manager / Team Leader, native mobile and responsive web (2014–2015). Newtech Sistemas (Venezuela), Development Consultant, .NET applications and WebServices (2006–2010).

HANDS-ON AI PROJECTS

Pulse AI (4Told FinTech): Production agentic AI platform: multi-agent orchestration, multi-engine LLMs, Drools inferential reasoning, RAG and SLM for regulated finance and insurance. Role: product and architecture direction.

AI Sommelier / Zona Barrica (personal): Built end-to-end on Google Cloud Vertex AI, a personalized recommendation engine using Gemini 2.5 Flash to generate real sensory profiles from user tasting data. Full-stack, React + FastAPI + Firestore. Around 100 initial users.

Runcoach AI (personal): AI-generated personalized running training plans, built on Google Cloud Vertex AI with Python, with prompt design, model configuration and generation pipeline implemented hands-on.

Digital Onboarding (4Told FinTech): Identity verification and anti-impersonation platform: biometric validation, OCR, document validation and electronic signature, orchestrated across third-party services under regulatory constraints.

Orienta & Loanware (Vedana Solutions): AI analytics over a data lake for enterprise decision-making.

EDUCATION & LANGUAGES

M.Sc. in Information and Communication Technologies at Universidad del Zulia (in progress, expected 2027)

B.Sc. in Computer Science at Universidad del Zulia (2007)

Spanish (native) · English (professional working proficiency, B2)